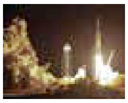


Hefty SpaceX rocket launched
SpaceX recently launched its heftiest rocket holding 24 research satellites. **https://dgit.in/july19-34**

Cookies in space
Double Tree by Hilton announces that the first space-baked cookies will be produced soon. **https://dgit.in/july19-35**

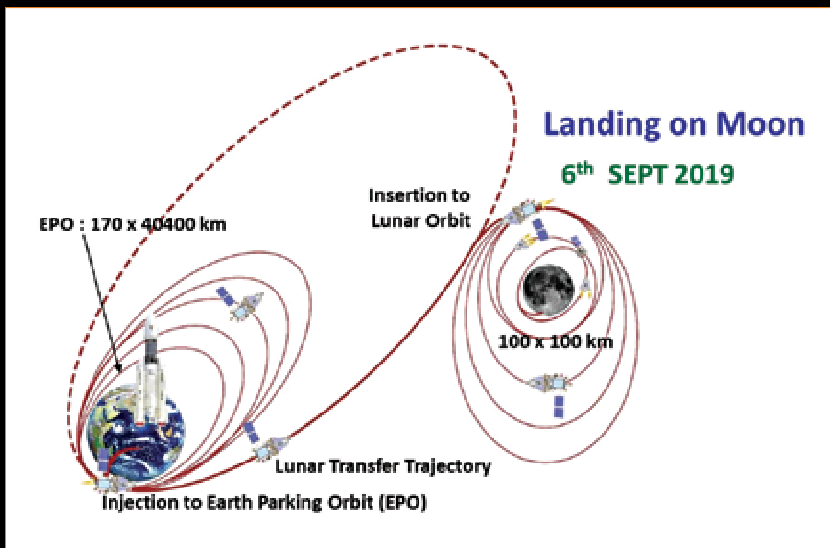
the south pole of the Moon, before it separates from the lander.

Then comes the really tricky part. The lander has to break away from the orbiter, and execute a soft landing on the Moon. It has to shed all its velocity in less than fifteen minutes. K Sivan divulged some thoughts about this stage of the operation and said, "these fifteen minutes are going to be the most terrifying for the people who belong to ISRO, probably for the scientists and the people in India also. This is going to be the most difficult task ever undertaken by ISRO in a mission." This is a technically a more advanced operation than the previous mission. The lander in Chandrayaan crashed into the Moon. The algorithms for executing the soft landing have been developed entirely in-house, so the landing itself is a demonstration of the technical prowess of Indian scientists. During the touchdown, the lander will extend its legs to dampen the landing, and protect the delicate electronic components on board. ISRO tested the operations of the lander at a special facility built near Chitradurga. If successful, the soft landing will be the first near the south pole of the Moon, and India will become the fourth country to execute such a landing. After the landing, the rover will be slowly deployed over the course of four hours. The rover will then explore the lunar surface. It has a maximum range of 500 meters. The rover is semi-autonomous, and therefore, will be receiving partial control commands from the ground station. ISRO created artificial lunar soil to test the rover and make sure that it works as expected.

PAYLOADS AND OBJECTIVES

At this point, the mission will have already achieved its primary goal, which is to demonstrate the capabilities of soft landing on the lunar surface, and operate a rover on the surface. However, on board the lander, rover and the orbiter are a number of scientific payloads, that will study the region.

The instruments that will facilitate all these experiments were designed, developed and made by various ISRO facilities located all across India. These include the Space Application Centre (SAC) in Ahmedabad, the Space Physics Laboratory (SPL) in Thiruvananthapuram, and the Space Astronomy Group (SAG) in Bengaluru. The Terrain Mapping Camera 2 (TMC2) on board the orbiter will take sub 5 m resolution images of the lunar surface, and will allow ISRO scientists to create 3D maps. The Chandrayaan 2 Large Area Soft X-Ray Spectrometer (CLASS) will examine the presence and distribution of rock forming elements such as Magnesium, Aluminium, Silicon, Calcium, Titanium, Iron, and Sodium. These elements will be identified based on the characteristic emissions when exposed to sunlight. The Solar X-Ray Monitor (SXM) will support the CLASS instrument, and it measures the intensity of solar radiation, which in turn, facilitates the accurate calibration of the observations and measurements from CLASS. The Orbiter High Resolution Camera (OHRC) will be used to identify the landing area, and generate 3D maps to ensure that the lander performs as expected. The Imaging Infrared Spectrometer (IIRS) will be used to map the distribution of minerals and finding water or hydroxyls. The Dual Frequency Synthetic Aperture Radar will be used to map the polar regions, measure the thickness of the regolith (dry soil) on the lunar surface and estimate the quantity of water ice in the polar regions. The Chandrayaan 2 Atmospheric Compositional Explorer 2 (CHACE2) will continue the measurements of the Chandrayaan mission, and study the rarified exosphere of the already tenuous lunar atmosphere. Finally, the Dual Frequency Radio Science (DFRS) experiment will measure the electron density in the ionosphere of the Moon. Two other instruments developed for the mission, the Chandrayaan Surface Thermophysical Experiment (ChaSTE) and the Radio Anatomy of Moon Bound Hyper-Sensitive Atmosphere and Ionosphere (RAMBHA) payloads, did not make it to the final payload.



Credit: ISRO

The path Chandrayaan will take to the Moon

Broadly speaking, all the three components of the mission hope to conduct topological and mineralogical studies of the lunar south pole. The data from these observations will be made available to the Indian scientific community. If the Chandrayaan mission is anything to go by, the observations by Chandrayaan 2 will continue to form the basis of new research long after the planned mission duration of one year. d